

CCA Project

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1 Resultant

For the entire section, let $f, g \in \mathbb{Z}[x]$ with $\deg f = n$ and $\deg g = m$.

First, we define the Sylvester matrix, $S(f, g) \in \mathbb{Z}^{(n+m) \times (n+m)}$, as follows [1] :

$$S(f, g) = \begin{pmatrix} f_n & & & & g_m & & & \\ f_{n-1} & f_n & & & g_{m-1} & g_m & & \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & \\ f_0 & f_1 & \cdots & f_n & g_0 & g_1 & \cdots & \\ & f_0 & \cdots & \vdots & & g_0 & \cdots & \\ & & \ddots & f_1 & & & \ddots & \\ & & & f_0 & & & & g_0 \end{pmatrix}$$

where all the non-specified coefficients are null. We can observe that the first m columns of $S(f, g)$ are coefficients of $f(x)$ shifted m times, and the last n columns are coefficients of $g(x)$ shifted n times. We can also define the matrix with reverse order of f and g , or by shifting in rows instead of columns.

Then, the resultant of f and g is defined as [1]:

$$\text{res}(f, g) = \det S(f, g)$$

The resultant is important, as we have the following property:

$$\text{res}(f, g) \neq 0 \iff \gcd(f, g) = 1$$

which implies that the resultant of two polynomials is 0 if and only if they have a common root.

2 Subresultants

As before, for the entire section, let $f, g \in \mathbb{Z}[x]$ with $\deg f = n$ and $\deg g = m$. Moreover, we assume $m \leq n$.

Resultant of polynomials is important, as it gives as an efficient manner to understand if two polynomials have a common root or not. However, it does not give any other information but about gcd. Thus, we can extend the resultant definition into subresultants, which include all results of the Extended Euclidean Algorithm [2].

First, we define Sylvester matrices for subresultants, similar to the resultant's matrix. Given $0 \leq k \leq m$,

$$S_k(f, g) = \begin{pmatrix} f_n & & & & g_m & & & \\ f_{n-1} & f_n & & & g_{m-1} & g_m & & \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & \\ f_{n-m+k+1} & \cdots & \cdots & f_n & g_{k+1} & \cdots & \cdots & g_m \\ \vdots & & & \vdots & \vdots & & \ddots & \\ f_{k+1} & \cdots & \cdots & f_m & g_{m-n+k+1} & \cdots & \cdots & g_m \\ \vdots & & & \vdots & \vdots & & & \vdots \\ f_{2k-m+1} & \cdots & \cdots & f_k & g_{2k-n+1} & \cdots & \cdots & g_k \end{pmatrix}$$

, which can be understand as removing the last k -columns of each submatrix of f and g , and then removing the last $2k$ rows of the matrix.

Then, the k -th subresultant of f and g is defined as [2]:

$$\alpha_k = \det S_k(f, g)$$

References

- [1] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra*. Cambridge University Press, Cambridge, 1999. Section 6.3: The resultant.
- [2] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra*. Cambridge University Press, Cambridge, 1999. Section 6.10: Subresultants.